

## SUMMARY TABLE: GENERAL SCIENCE GRADE 10

### Sub-Strand 1.7: Microorganisms — Teacher Reference (Pre-filled)

| SUMMARY TABLE: GENERAL SCIENCE GRADE 10 |                                                                                                                                                                                                |
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| Sub-Strand                              | Sub-Strand 1.7: Microorganisms                                                                                                                                                                 |
| Driving Question                        | DRIVING QUESTION: Why did the ugali and chapati grow fuzzy patches overnight — and how can understanding the invisible organisms responsible help us protect our health and our food in Kenya? |

### INSTRUCTIONS

FOR TEACHERS: This is the teacher reference version — each row is pre-filled with expected student responses. Use it to assess student Summary Tables, identify gaps, and prepare discussion questions. The student version has blank cells for students to complete after each lesson.

| Lesson # and Title                                                                                      | What did I observe?                                                                                                                                                                                                                                                             | What did I learn?                                                                                                                                                                                                                                                                                                                                                            | How does this explain the phenomenon?                                                                                                                                                                                                                                                                                                                                                                                        |
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| <b>Lesson 1</b><br><b>Noticing and Wondering:</b><br><b>What Is Growing on Our Ugali and Chapati?</b>   | Photographs of leftover ugali and chapati stored overnight on a kitchen counter in Kisumu, showing white, green and black fuzzy patches, slimy areas and described sour smell. Students observed variation in patch colour, texture and location across the same food surfaces. | Students surfaced prior knowledge about tiny organisms ('germs', 'bacteria') and began to distinguish between living and non-living causes of food change. They learned that scientists separate observations from inferences, and that the patches — whatever they are — appeared without anyone touching the food, meaning the source must be investigated systematically. | No formal explanation was provided in Lesson 1 — this is deliberate. Students' initial mental models (Model 1) record their pre-instruction thinking. The Driving Question Board now holds the class's collective questions, organised into four clusters, which will be answered across Lessons 2–6 as students learn about types of microorganisms, modes of transmission, infections, prevention and economic importance. |
| <b>Lesson 2</b><br><b>Types of Microorganisms:</b><br><b>Naming the Invisible Invaders on Our Ugali</b> | We observed photomicrographs and illustrations of fungi, bacteria, and viruses, and we examined photographs of the white, green, and black fuzzy patches growing on the ugali and chapati from our phenomenon.                                                                  | We learned that there are three major types of microorganisms — fungi (eukaryotic, with chitin cell walls), bacteria (prokaryotic, no true nucleus), and viruses (non-cellular, the smallest of the three) — and that the fuzzy patches on the ugali and chapati are caused by fungal moulds whose spores were already present in the kitchen air.                           | We can now explain that the patches appeared without anyone touching the food because microscopic fungal spores, invisible to the naked eye, were floating in the kitchen air and settled on the warm, moist, uncovered ugali and chapati overnight, where they germinated and grew into the visible fuzzy colonies we see in the phenomenon photographs.                                                                    |
| <b>Lesson 3</b><br><b>Modes of Transmission:</b><br><b>How Did the Mould Get onto the Ugali?</b>        | Students researched five modes of transmission (airborne, direct contact, respiratory droplets, contaminated water/food, vectors) using digital or print                                                                                                                        | Airborne transmission — the movement of fungal spores suspended in air currents — is the specific mechanism that best explains how mould appeared on the uncovered                                                                                                                                                                                                           | Students constructed a Cause-and-Effect Chain tracing the path of mould spores from an environmental source, through the air, onto the nutrient-rich surface of warm ugali                                                                                                                                                                                                                                                   |

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|                                                                                                | media resource packs, presented findings in a Gallery Walk format, and gathered evidence on how microorganisms travel from source to host in Kenyan contexts.                                                                                                                                                                                                                                                                                                                                                                                                                                                      | ugali and chapati in the Kisumu kitchen overnight. Different transmission routes determine which prevention and control methods are most effective.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | left uncovered in a Kisumu kitchen, producing the white, green, and black fuzzy patches and sour smell observed in the anchoring phenomenon.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| <b>Lesson 4</b><br><b>Infections Caused by Microorganisms: Fungi, Bacteria, and Viruses</b>    | Learners examined ARES information cards showing photographs and descriptions of fungal infections (ringworm on a child's arm in Kisumu; oral thrush patches), bacterial infections ( <i>Vibrio cholerae</i> linked to Lake Victoria water sources; <i>Salmonella typhi</i> ; <i>Mycobacterium tuberculosis</i> in Nairobi informal settlements), and viral infections (influenza, HIV, COVID-19 in the Kenyan context). They gathered evidence through a Jigsaw activity, completing a four-column Evidence Table that recorded infection name, causative organism, mode of transmission, and body site affected. | Learners learned that microorganisms causing human infections fall into three categories — fungi (same kingdom as the mould on the ugali and chapati), bacteria, and viruses — each with distinct causative organisms, transmission routes, and body sites affected. They learned that the mode of transmission identified in Lesson 3 (airborne, waterborne, direct contact, bodily fluids) directly predicts the type of infection that results, and that Kenyan communities near Lake Victoria face particular bacterial infection risks due to waterborne transmission of <i>Vibrio cholerae</i> .                                                                                                                                                                                                           | Learners constructed explanations linking the mould patches on the ugali and chapati to the broader world of microorganism-caused infections: the same fungal kingdom that spoils food overnight can infect human skin and mucous membranes; bacteria in contaminated water cause cholera and typhoid; viruses spread through droplets and bodily fluids to cause influenza, COVID-19, and HIV. These explanations were recorded in annotated body-outline models and three-sentence 'Explanation So Far' notebook entries, and were used to revise and extend the Driving Question Board.                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| <b>Lesson 5</b><br><b>Prevention and Control: How Mode of Transmission Shapes Our Defences</b> | Three Kisumu household vignettes (Mama Adhiambo in Nyalenda, Mzee Otieno in Siaya, the Kamau household in Milimani) showing different food-storage realities, infection histories, and available resources; photographs of covered versus uncovered versus refrigerated ugali; a Transmission-to-Control Matching Table completed in groups; peer prevention posters.                                                                                                                                                                                                                                              | The mode of transmission of a microorganism determines which control measure is most effective: covering food interrupts airborne/droplet deposition of spores (fungal and bacterial); hand-washing interrupts direct contact transmission; boiling water and cooking food thoroughly interrupts the contaminated food/water route; vector control and sanitation interrupt vector-borne and faecal-oral routes. Not all scientifically valid controls are equally accessible — low-income households in rural Kenya (e.g., Siaya County) and urban informal settlements (e.g., Nyalenda, Kisumu) may lack refrigeration, running water, or sealed storage, requiring community- and systems-level alternatives such as women's group sensitisation, school hygiene clubs, and low-cost food-covering practices. | The overnight growth of white, green, and black fuzzy patches on the ugali and chapati in the Kisumu kitchen occurred because the uncovered food surface provided an unobstructed landing site for airborne fungal spores (and bacterial cells) already present in the kitchen environment. Covering the sufuria — even with a simple wire mesh dome or a clean cloth, which costs nothing — would have physically blocked the airborne transmission route and dramatically reduced microbial colonisation. Refrigeration would additionally have slowed the germination and growth of any spores that did land, by reducing the temperature below the optimum growth range of most mesophilic fungi and bacteria. The variation in patch colour (white, green, black) across the same food surface reflects colonisation by DIFFERENT species of mould (e.g., <i>Rhizopus</i> , <i>Penicillium</i> , <i>Aspergillus</i> ), each arriving via the same airborne route but occupying slightly different microhabitats on the food substrate. |
| <b>Lesson 6</b>                                                                                | Learners observed a juxtaposition of the                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Learners learned that microorganisms have                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | Learners explained, through a co-                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |

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| <b>Economic Importance of Microorganisms: The Dual Nature of the Invisible World</b>                                                 | <p>spoiled ugali/chapati phenomenon photograph alongside images of Kenyan commercial yoghurt, mandazi, and maize flour — products linked to microorganism activity. They also observed peer-generated sticky notes on a Benefit–Harm wall chart produced during the jigsaw expert group activity, and a classmate's annotated cumulative model projected for whole-class analysis.</p>                    | <p>a dual economic nature: (1) Beneficial roles — <i>Lactobacillus</i> ferments milk/uji, <i>Saccharomyces</i> leavens bread and mandazi, <i>Rhizobium</i> fixes atmospheric nitrogen in Kenyan bean and maize farms, <i>Penicillium</i> produces penicillin antibiotics, methane-producing bacteria generate biogas for rural Kenyan households, and decomposer bacteria/fungi recycle nutrients to maintain soil fertility; (2) Harmful roles — fungal and bacterial spoilage of food (directly causing the ugali/chapati phenomenon) generates significant post-harvest economic losses for Kenyan market vendors, and fungal pathogens cause crop diseases such as maize rust and coffee berry disease. The same biological kingdom (e.g., fungi) contains both spoilage agents and commercially vital organisms.</p> | <p>constructed T-chart concept organiser, a written 'Explain It to a Market Vendor' paragraph, and an updated cumulative model, that the microorganisms responsible for the fuzzy patches on the Kisumu family's ugali and chapati belong to the same broad taxonomic groups as organisms that underpin critical sectors of Kenya's food economy, agricultural productivity, and healthcare system. The context-dependence of microorganism impact — harmful when growing unchecked on stored food, beneficial when harnessed in controlled fermentation, farming, or pharmaceutical processes — was articulated as the central explanatory principle. Learners updated the Driving Question Board to reflect this dual-nature understanding and identified open questions about prevention and control to be addressed in Lesson 7.</p> |
| <b>Lesson 7</b><br><b>Project Investigation: Mould Growth on Food Substrates — What Conditions Do Microorganisms Need to Thrive?</b> | <p>Students observed that pre-incubated mould samples on warm moist uncovered ugali and bread showed extensive white, green and black fuzzy growth, while dry cool covered samples showed little or no visible growth. They recorded colour, texture, estimated coverage and smell across six container conditions and noted that the dry maize biscuit showed the least growth under all conditions.</p> | <p>Students learned that mould growth is favoured by three key conditions: warmth (which speeds up fungal enzyme activity and hyphal growth), moisture (which provides the water needed for enzymatic digestion and nutrient absorption), and exposure to air (which allows airborne spores to settle and germinate on the food surface). Reducing any one of these conditions slows or prevents spoilage, which is why covering food, refrigerating it, or drying it are effective preservation strategies.</p>                                                                                                                                                                                                                                                                                                          | <p>The investigation explained the anchoring phenomenon directly: the ugali and chapati in Kisumu spoiled overnight because the kitchen was warm, the food was moist, and it was left uncovered — all three conditions that maximise mould spore germination and hyphal growth. It also explained traditional Kenyan food preservation practices such as drying omena and salting meat as empirically effective strategies that reduce water availability and temperature, thereby preventing microbial growth without requiring modern refrigeration.</p>                                                                                                                                                                                                                                                                               |
| <b>Lesson 8</b><br><b>Model Building, DQB Synthesis and Community Sensitisation</b>                                                  | <p>Students observed their own revised model sheets alongside pre-incubated mould samples from Lesson 7, the full Driving Question Board with all accumulated questions and predictions from eight lessons, and three group model presentations — all anchored to the original photographs of spoiled ugali and chapati from the Kisumu kitchen.</p>                                                      | <p>Students confirmed that the fuzzy patches on the ugali and chapati were fungal mould colonies that grew from airborne spores which landed on uncovered, warm and moist food overnight; that bacteria and viruses also cause disease through specific transmission routes; that covering food, refrigerating it or drying it prevents spoilage by removing conditions microorganisms need to grow; and that microorganisms also play beneficial roles in Kenyan food production, agriculture and medicine.</p>                                                                                                                                                                                                                                                                                                          | <p>Students explained the full causal chain of the phenomenon — from invisible airborne fungal spores dispersing through the kitchen air, settling on uncovered ugali and chapati, germinating in warm moist conditions overnight, forming visible white, green and black mould colonies — and communicated evidence-based food preservation advice to specific community audiences in Kenya, demonstrating that science understanding can protect health and economic wellbeing at the household</p>                                                                                                                                                                                                                                                                                                                                    |

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ATTRIBUTION AND LICENSING

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**Content & Pedagogy:** Developed and adapted by the educator team at SeaVuria (registered 501(c)(3) in the United States and registered NGO in Kenya, No. 872-850A-BF11), who designed the pedagogical framework, lesson templates and exemplar lessons on which this material is based.

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